

Last time Sequential Criterion for Limits

If (a_n) is a sequence s.t. $a_n \neq a$ for almost all n and $\lim_{n \rightarrow \infty} a_n = a$
and $\lim_{n \rightarrow \infty} f(a_n) = l$ for all such sequences (a_n)

then $\lim_{x \rightarrow a} f(x) = l$.

Pf. Suppose $\lim_{x \rightarrow a} f(x) \neq l$ even though $\lim_{n \rightarrow \infty} f(a_n) = l$

Goal Construct a seq $(a_n) \rightarrow a$ but $f(a_n) \not\rightarrow l$

Since $\lim_{x \rightarrow a} f(x) \neq l$

$\exists \varepsilon$ s.t. for any $\delta > 0$

$\exists x$ with $0 < |x - a| < \delta$, $|f(x) - l| < \varepsilon$

To find a_1 : let $\delta = 1$

then $\exists x$ as above, so let $a_1 = x$

To find a_2 : let $\delta = \frac{1}{2}$

then $\exists x$ as above, so let $a_2 = x$

To find a_n : let $\delta = \frac{1}{n}$

then $\exists x$ as above, $a_n = x$

Then we've constructed a seq $(a_n) \rightarrow a$ but $|f(a_n) - l| < \varepsilon$

so $\lim_{n \rightarrow \infty} f(a_n) = l$

This violates our assumption.



Def. A seq (a_n) is strictly increasing/monotone if $a_{n+1} \geq a_n \forall n \in \mathbb{Z}_{>0}$

It is decreasing/monotone if $a_{n+1} \leq a_n \forall n \in \mathbb{Z}_{>0}$

A sequence is bounded above if $\exists B$ s.t. $a_n \leq B \forall n$.

Thm. Monotone Convergence Thm

If (a_n) is increasing + bounded above

$\Rightarrow (a_n)$ converges.

PF. Let $A = \{a_n : n > 0\}$.

• $A \neq \emptyset$ b/c. $a_n \in A \forall n$

• A bounded b/c. (a_n) is bounded

$\Rightarrow \alpha := \text{LUB } A$

claim $\lim_{n \rightarrow \infty} a_n = \alpha$

Let $\varepsilon > 0$, want N s.t. whenever $n > N$, then $|\alpha - a_n| < \varepsilon$

Suppose for a contradiction that such an N DNE.

$\exists n$ large enough s.t. $|\alpha - a_n| > \varepsilon$

First claim that $\exists a_n$ w/ $|\alpha - a_n| < \varepsilon$

If such an n DNE, then $\forall n |\alpha - a_n| = \alpha - a_n \geq \varepsilon$

$(\alpha - \varepsilon \geq a_n \forall n \Rightarrow \alpha - \varepsilon$ is an UB for A but it's smaller than α)

Let $N = n_0$

Then for $\forall n \geq N = n_0$

$\varepsilon > |\alpha - a_{n_0}| = \alpha - a_{n_0}$

$\geq \alpha - a_n = |\alpha - a_n|$

So this is the N we wanted

and $\lim_{n \rightarrow \infty} a_n = \alpha$



Lemma Any seq (a_n) contains a Subsequence that is monotone.
 $\rightarrow a_1, a_2, a_3, \dots$ is a subseq

Given (a_n) , an element a_n is a peak if it is larger than all numbers following it.

Case 1 If there are infinitely many peaks, take the subseq of peaks, which is decreasing

Case 2 If there are finitely many peaks.

WLOG assume that there are no peaks

(If there were any, we'll construct the subseq after all the peaks)

let $b_1 := a_1$

Since a_1 is not a peak, $\exists a_{n_2} > a_1$

let $b_2 := a_{n_2}$ ($b_2 > b_1$)

Since b_2 is not a peak, $\exists a_{n_3} > b_2$

let $b_3 := a_{n_3}$ etc $\Rightarrow (b_n)$ is an increasing subseq.



Thm. Bolzano - Weierstrass

Any bounded seq (a_n) has a convergent subseq.

Pf. By the previous lemma, (a_n) has a monotone subseq.

Since (a_n) bounded, so is the subseq $\Rightarrow$ this subseq converges.



Sequential Criterion for Boundedness

Let $A \neq \emptyset \subseteq \mathbb{R}$ be a set for which every seq w/ elements in A has a convergent subseq.

Then A is bounded.